

Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

1 Problem 1: Evaluate $\int_{-1}^1 \frac{\sqrt{1-x^2}}{x^2+1} dx$ by contour integration

Hint. Consider $F(z) = \sqrt{z^2-1}/(z^2+1)$ with a branch cut on $[-1, 1]$, and integrate around a large circle.

Background

Fix the branch of $G(z) = \sqrt{z^2-1}$ with cut on $[-1, 1]$ and $G(x) > 0$ for $x > 1$. For $x \in (-1, 1)$,

$$G(x+i0) = +i\sqrt{1-x^2}, \quad G(x-i0) = -i\sqrt{1-x^2}.$$

Hence the sum of the upper and lower lip integrals across the cut equals

$$2i \int_{-1}^1 \frac{\sqrt{1-x^2}}{x^2+1} dx = 2i I.$$

Solution

Define

$$F(z) = \frac{\sqrt{z^2-1}}{z^2+1}, \quad z \in \mathbb{C} \setminus [-1, 1].$$

Poles: $z = \pm i$ (simple). Large-circle behavior: $F(z) = \frac{1}{z} + O(z^{-3})$; thus

$$\int_{|z|=R} F(z) dz \longrightarrow 2\pi i \quad (R \rightarrow \infty).$$

Deforming the big circle to a contour hugging the cut gives

$$\underbrace{\int_{|z|=R} F(z) dz}_{\rightarrow 2\pi i} = \underbrace{\int_{\text{upper lip}} F dz + \int_{\text{lower lip}} F dz}_{= 2i I} + (\text{vanishing end caps}).$$

Residues at the simple poles:

$$\begin{aligned} \text{Res}_{z=i} F &= \lim_{z \rightarrow i} \frac{(z-i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{i^2-1}}{2i} = \frac{\sqrt{-2}}{2i} = \frac{\sqrt{2}}{2}, \\ \text{Res}_{z=-i} F &= \lim_{z \rightarrow -i} \frac{(z+i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{(-i)^2-1}}{-2i} = \frac{\sqrt{-2}}{-2i} = \frac{\sqrt{2}}{2}. \end{aligned}$$

Hence, by the residue theorem,

$$\int_{|z|=R} F(z) dz \longrightarrow 2\pi i \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) = 2\pi i \sqrt{2}.$$

Equating the two expressions for the big-circle integral,

$$2i I = 2\pi i (\sqrt{2} - 1) \quad \Rightarrow \quad \boxed{I = \pi(\sqrt{2} - 1)}.$$

Check (real calculus)

With $x = \sin \theta$,

$$I = 2 \int_0^{\pi/2} \frac{\cos^2 \theta}{1 + \sin^2 \theta} d\theta = 2 \int_0^\infty \frac{dt}{(1+t^2)(1+2t^2)} dt = \pi(\sqrt{2} - 1).$$

2 Problem 2: Fourier transforms and inverse verification

We use the convention $\widehat{f}(\xi) = \int_{-\infty}^\infty f(x) e^{-i\xi x} dx$ and $f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty \widehat{f}(\xi) e^{i\xi x} d\xi$.

(a) $f(x) = e^{-|x|}$ — Transform and step-by-step inverse

Transform:

$$\widehat{f}(\xi) = \int_0^\infty e^{-(1+i\xi)x} dx + \int_{-\infty}^0 e^{(1-i\xi)x} dx = \frac{1}{1+i\xi} + \frac{1}{1-i\xi} = \boxed{\frac{2}{1+\xi^2}}.$$

Inverse by residues (two cases):

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2e^{i\xi x}}{1+\xi^2} d\xi = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi.$$

$$\begin{aligned} \text{Case } x > 0: \quad f(x) &= \frac{1}{2\pi} \oint_{\Gamma^+} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi && \text{(close upward; Jordan's lemma)} \\ &= \frac{1}{2\pi} (2\pi i) \operatorname{Res}_{\xi=i} \left(\frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} \right) = \frac{i 2e^{ix}}{2i(i+i)} = e^{-x}. \end{aligned}$$

$$\begin{aligned} \text{Case } x < 0: \quad f(x) &= \frac{1}{2\pi} \oint_{\Gamma^-} \frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} d\xi && \text{(close downward)} \\ &= \frac{1}{2\pi} (2\pi i) \operatorname{Res}_{\xi=-i} \left(\frac{2e^{i\xi x}}{(\xi-i)(\xi+i)} \right) = e^x. \end{aligned}$$

Therefore $\boxed{f(x) = e^{-|x|}}$.

(b) $f(x) = e^{-a^2 x^2}$ ($a > 0$) — Transform and step-by-step inverse

Transform:

$$\begin{aligned} \widehat{f}(\xi) &= \int_{\mathbb{R}} e^{-a^2 x^2 - i\xi x} dx = e^{-\xi^2/(4a^2)} \int_{\mathbb{R}} e^{-a^2 \left(x + i\frac{\xi}{2a^2}\right)^2} dx \\ &\Rightarrow \boxed{\widehat{f}(\xi) = \frac{\sqrt{\pi}}{a} e^{-\xi^2/(4a^2)}}. \end{aligned}$$

(Justify by integrating over a large rectangle whose vertical sides are the real line and the line shifted up by $i\xi/(2a^2)$; the horizontal sides vanish by Gaussian decay.)

Inverse by a rectangular shift (explicit steps):

$$\begin{aligned}
f(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{\sqrt{\pi}}{a} e^{-\xi^2/(4a^2)} e^{i\xi x} d\xi \\
&= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} \int_{\mathbb{R}} \exp\left(-\frac{1}{4a^2}(\xi - 2ia^2x)^2\right) \exp(-a^2x^2) d\xi \quad (\text{complete the square}) \\
&= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \int_{\mathbb{R}+2ia^2x} \exp\left(-\frac{u^2}{4a^2}\right) du \quad (\text{shift contour vertically}) \\
&= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \int_{\mathbb{R}} \exp\left(-\frac{u^2}{4a^2}\right) du \quad (\text{no poles; horizontal sides vanish}) \\
&= \frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{a} e^{-a^2x^2} \cdot 2a\sqrt{\pi} = e^{-a^2x^2}.
\end{aligned}$$

3 Problem 3: Schwarz–Christoffel map from the half-plane to a rectangle

Statement

Write down, as an integral, the Schwarz–Christoffel map from the upper half-plane $\mathbb{H}$ to a rectangle, with vertices being the images of the points $z = \pm 1$ and $z = \pm a$ (with $a > 1$ real). Explain why a cannot be chosen arbitrarily, but is determined by the rectangle’s aspect ratio.

Background

For prevertices $x_k \in \mathbb{R}$ mapping to interior angles $\alpha_k\pi$ in the polygon,

$$f'(z) = C \prod_k (z - x_k)^{\alpha_k - 1} \quad (\text{up to an affine change of variables}).$$

A rectangle has $\alpha_k = \frac{1}{2}$ at each vertex, so every factor’s exponent is $-\frac{1}{2}$.

Solution

Place prevertices at $-a, -1, 1, a$ (in order) with $a > 1$. Then

$$f'(z) = C (z+a)^{-1/2} (z+1)^{-1/2} (z-1)^{-1/2} (z-a)^{-1/2} = \frac{C}{\sqrt{(z^2-1)(z^2-a^2)}}.$$

Integrating,

$$f(z) = A + C \int^z \frac{dt}{\sqrt{(t^2-1)(t^2-a^2)}}$$

maps $\mathbb{H}$ conformally to a rectangle (up to translation and scaling).

Aspect ratio determines a

Side lengths are proportional to real integrals between consecutive prevertices:

$$W(a) = \int_{-1}^1 \frac{dt}{\sqrt{(1-t^2)(a^2-t^2)}}, \quad H(a) = \int_1^a \frac{dt}{\sqrt{(t^2-1)(a^2-t^2)}}.$$

Let $k = 1/a \in (0, 1)$. Standard substitutions give

$$W(a) = \frac{2}{a} K(k), \quad H(a) = \frac{2}{a} K'(k),$$

where K is the complete elliptic integral of the first kind and $K'(k) = K(\sqrt{1-k^2})$. Thus

$$\frac{H(a)}{W(a)} = \frac{K'(k)}{K(k)} \quad \text{with } k = \frac{1}{a}.$$

Given a rectangle, this ratio fixes k (hence a) uniquely; therefore a is not free.

4(a). Statement and branch points

Suppose $\zeta^2 = z^2 + 1$ and

$$z - i = r_1 e^{i\theta_1}, \quad z + i = r_2 e^{i\theta_2} \quad (r_1, r_2 > 0).$$

Then

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}}.$$

Reason. $z^2 + 1 = (z - i)(z + i) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$. Any square root has the two signs above. The branch points of the multifunction $(z^2 + 1)^{1/2}$ are where either factor vanishes, i.e. $z = \pm i$.

4(b). A branch with a finite vertical cut

Define the branch by

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}.$$

Values across the imaginary axis

Let $z = x + iy$ and set $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$. Note

$$r_1 = |y - 1|, \quad r_2 = |y + 1|, \quad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i) $|y| < 1$. $y - 1 < 0$, $y + 1 > 0$.

$$\begin{aligned} x \rightarrow 0^+ : \theta_1 \rightarrow (-\pi/2)^+, \theta_2 \rightarrow (\pi/2)^- &\Rightarrow \phi(y) \rightarrow 0, \\ x \rightarrow 0^- : \theta_1 \rightarrow (3\pi/2)^-, \theta_2 \rightarrow (\pi/2)^+ &\Rightarrow \phi(y) \rightarrow \pi. \end{aligned}$$

Hence

$$f(0^+ + iy) = \sqrt{1 - y^2}, \quad f(0^- + iy) = -\sqrt{1 - y^2}.$$

(ii) $y > 1$. $\theta_1 = \theta_2 = \pi/2$ on both sides, so $\phi(y) = \pi/2$ and

$$f(\pm 0 + iy) = i\sqrt{y^2 - 1}.$$

(iii) $y < -1$. $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$, so $\phi(y) = -\pi/2$ and

$$f(\pm 0 + iy) = -i\sqrt{y^2 - 1}.$$

Branch cut and behaviour at infinity

The only jump occurs for $|y| < 1$ on the imaginary axis, hence the branch cut is

$$S = \{x + iy : x = 0, |y| \leq 1\}.$$

For $|z| \rightarrow \infty$,

$$f(z) = |z|e^{i \arg z} (1 + o(1)) = z(1 + o(1)),$$

so $f(z) \sim z$. The image of S is the real segment $[-1, 1]$; the imaginary axis outside S maps to the imaginary axis. Thus $f : \mathbb{C} \setminus S \rightarrow \mathbb{C} \setminus [-1, 1]$ is conformal with $f(z) \sim z$.

4(c). A branch with two semi-infinite vertical cuts

Now define

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}, \quad -\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}.$$

Values across the imaginary axis

Let $z = x + iy$, $x \rightarrow 0^\pm$.

- $|y| < 1$: we may keep $\theta_1 \approx -\pi/2$, $\theta_2 \approx \pi/2$ on both sides, hence

$$f(\pm 0 + iy) = \sqrt{1 - y^2} \quad (\text{no jump}).$$

- $y > 1$: crossing the axis forces θ_1 to jump by 2π ; thus

$$f(0^+ + iy) = +i\sqrt{y^2 - 1}, \quad f(0^- + iy) = -i\sqrt{y^2 - 1}.$$

- $y < -1$: analogously,

$$f(0^+ + iy) = -i\sqrt{y^2 - 1}, \quad f(0^- + iy) = +i\sqrt{y^2 - 1}.$$

Therefore the branch cut consists of the two rays $(-\infty i, -i] \cup [i, +\infty i)$.

Values on the real axis and asymptotics by half-planes

For $z = x \in \mathbb{R}$,

$$r_1 = r_2 = \sqrt{x^2 + 1}, \quad \frac{\theta_1 + \theta_2}{2} = \begin{cases} 0, & x > 0, \\ \pi, & x < 0. \end{cases}$$

Hence

$$\boxed{f(x) = \operatorname{sgn}(x) \sqrt{x^2 + 1}}.$$

As $|z| \rightarrow \infty$,

$$f(z) \sim |z| e^{i \arg z} \times \begin{cases} 1, & \Re z > 0, \\ -1, & \Re z < 0, \end{cases}$$

i.e.

$$\boxed{f(z) \sim z \text{ for } \Re z > 0, \quad f(z) \sim -z \text{ for } \Re z < 0.}$$

Exact problem statement (as in the text)

4. Branches of $\sqrt{z^2 + 1}$ — Problem statement

(a) **(a)** Show that, if $\zeta^2 = z^2 + 1$ and

$$z = i + r_1 e^{i\theta_1} = -i + r_2 e^{i\theta_2}, \quad r_1, r_2 > 0, \quad \theta_1, \theta_2 \in \mathbb{R},$$

then

$$\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}.$$

Explain briefly why $z = \pm i$ are the branch points of the multifunction $(z^2 + 1)^{1/2}$.

(b) **(b)** Consider the branch defined by

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}.$$

State the values of $(\theta_1 + \theta_2)/2$ on either side of the imaginary axis, and hence compute $f(\pm i)$ in terms of y for $y \in \mathbb{R}$. Deduce that the branch cut is

$$S = \{x + iy : x = 0, |y| \leq 1\}.$$

Show that $f(z) \sim z$ as $|z| \rightarrow \infty$ (i.e. $f(z)/z \rightarrow 1$ as $|z| \rightarrow \infty$). Sketch the image of the cut z -plane $\mathbb{C} \setminus S$ under the map $\zeta = f(z)$.

(c) **(c)** Consider instead the branch

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}, \quad -\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}, \quad -\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}.$$

State the values of $(\theta_1 + \theta_2)/2$ on either side of the imaginary axis, and hence compute $f(\pm i)$ in terms of y for $y \in \mathbb{R}$. Deduce that the branch cut is along the imaginary axis from $z = -i\infty$ to $z = -i$ and from $z = i$ to $z = i\infty$. Compute $f(x)$ in terms of x for $x \in \mathbb{R}$. Show that

$$f(z) \sim \begin{cases} z, & \text{as } |z| \rightarrow \infty \text{ with } \Re(z) > 0, \\ -z, & \text{as } |z| \rightarrow \infty \text{ with } \Re(z) < 0. \end{cases}$$

Sketch the images of the half-planes $\Re(z) > 0$ and $\Re(z) < 0$ under the map $\zeta = f(z)$.

Solutions (with background)

Throughout, write

$$z - i = r_1 e^{i\theta_1}, \quad z + i = r_2 e^{i\theta_2}, \quad r_1, r_2 > 0,$$

so that

$$z^2 + 1 = (z - i)(z + i) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

(a) Formula for ζ and branch points

Since $\zeta^2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$, any square root is

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}}.$$

Branch points occur at zeros of $z^2 + 1$, where an argument for a factor cannot be chosen continuously: these are exactly $z = \pm i$.

(b) Branch with finite vertical cut $S = \{x = 0, |y| \leq 1\}$

Branch choice. Take $-\frac{\pi}{2} < \theta_1, \theta_2 \leq \frac{3\pi}{2}$ and define $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}$.

Angles on the imaginary axis. Let $z = x + iy$ and denote $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$. Use

$$r_1 = |y - 1|, \quad r_2 = |y + 1|, \quad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i) $|y| < 1$. As $x \rightarrow 0^+$: $\theta_1 \rightarrow (-\pi/2)^+$, $\theta_2 \rightarrow (\pi/2)^- \Rightarrow \phi(y) \rightarrow 0$. As $x \rightarrow 0^-$: to keep θ_1 in $(-\pi/2, 3\pi/2]$ we add 2π , so $\theta_1 \rightarrow (3\pi/2)^-$, $\theta_2 \rightarrow (\pi/2)^+ \Rightarrow \phi(y) \rightarrow \pi$. Hence

$$\boxed{f(0^+ + iy) = \sqrt{1 - y^2}, \quad f(0^- + iy) = -\sqrt{1 - y^2} \quad (|y| < 1)}.$$

(ii) $y > 1$. $\theta_1 = \theta_2 = \pi/2$ from either side, so

$$\boxed{f(\pm 0 + iy) = i\sqrt{y^2 - 1} \quad (y > 1)}.$$

(iii) $y < -1$. $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$, so

$$\boxed{f(\pm 0 + iy) = -i\sqrt{y^2 - 1} \quad (y < -1)}.$$

Branch cut. The only sign jump occurs for $|y| < 1$ on the imaginary axis, hence

$$\boxed{S = \{x + iy : x = 0, |y| \leq 1\}}.$$

Asymptotics. As $|z| \rightarrow \infty$,

$$r_1 r_2 = |z - i||z + i| = |z^2 + 1| = |z|^2(1 + o(1)), \quad \theta_1 + \theta_2 = \arg(z^2 + 1) = 2 \arg z + o(1),$$

so

$$\boxed{f(z) \sim z \quad (|z| \rightarrow \infty)}.$$

Image sketch. The cut segment S maps to the real interval $[-1, 1]$ (opposite sides map to $(0, 1]$ and $[-1, 0)$); outside S the imaginary axis maps to the imaginary axis. Thus $f : \mathbb{C} \setminus S \rightarrow \mathbb{C} \setminus [-1, 1]$ is conformal with $f(z) \sim z$.

(c) Branch with two semi-infinite cuts

Branch choice. Take $-\frac{3\pi}{2} < \theta_1 \leq \frac{\pi}{2}$ and $-\frac{\pi}{2} < \theta_2 \leq \frac{3\pi}{2}$, and define $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1 + \theta_2)/2}$.

Angles across the imaginary axis. For $z = x + iy$:

(i) $|y| < 1$. We may keep $\theta_1 \approx -\pi/2$ and $\theta_2 \approx \pi/2$ on both sides, hence

$$\boxed{f(\pm 0 + iy) = \sqrt{1 - y^2} \quad (|y| < 1)}.$$

(ii) $y > 1$. Crossing the axis forces θ_1 to jump by 2π ; therefore

$$\boxed{f(0^+ + iy) = +i\sqrt{y^2 - 1}, \quad f(0^- + iy) = -i\sqrt{y^2 - 1} \quad (y > 1)}.$$

(iii) $y < -1$. Similarly,

$$\boxed{f(0^+ + iy) = -i\sqrt{y^2 - 1}, \quad f(0^- + iy) = +i\sqrt{y^2 - 1} \quad (y < -1)}.$$

Branch cuts. The discontinuities occur on the rays

$$\boxed{(-i\infty, -i] \cup [i, +i\infty)}.$$

Values on the real axis. For $z = x \in \mathbb{R}$, $r_1 = r_2 = \sqrt{x^2 + 1}$ and

$$\frac{\theta_1 + \theta_2}{2} = \begin{cases} 0, & x > 0, \\ \pi, & x < 0. \end{cases} \quad \Rightarrow \quad \boxed{f(x) = \operatorname{sgn}(x) \sqrt{x^2 + 1}}.$$

Asymptotics by half-planes. As $|z| \rightarrow \infty$,

$$\boxed{f(z) \sim z \text{ for } \Re z > 0, \quad f(z) \sim -z \text{ for } \Re z < 0}.$$

4 Problem 5: Images under $\zeta = \frac{1}{z}$, $\zeta = \sin z$, and $\zeta = z + e^z$

(a) **Intersection of $|z - 1| < 1$ and $|z + i| < 1$ under $\zeta = 1/z$**

Background (inversion). If $w = 1/z$ and $|z - a| = |a|$ (a circle through 0), then

$$|1 - aw| = |a||w| \iff 1 = 2\Re(aw) \iff \Re(aw) = \frac{1}{2},$$

i.e. the image is a straight line.

Boundary images.

$$|z - 1| = 1 \mapsto \Re(w) = \frac{1}{2}, \quad |z + i| = 1 \mapsto \Re((-i)w) = \frac{1}{2} \iff \Im(w) = \frac{1}{2}.$$

Which side is the interior?

$$|z - 1| < 1 \iff |1/w - 1| < 1 \iff |1 - w| < |w| \iff \Re(w) > \frac{1}{2};$$

$$|z + i| < 1 \iff |1 + iw| < |w| \iff \Im(w) > \frac{1}{2}.$$

Image. The lens $|z - 1| < 1 \cap |z + i| < 1$ maps to the quarter-plane

$$\boxed{\{w : \Re w > \frac{1}{2}, \Im w > \frac{1}{2}\}}.$$

(b) Strip $-\frac{\pi}{2} < x < \frac{\pi}{2}$ **under** $\zeta = \sin z$

Write $z = x + iy$. Then

$$\sin z = \sin x \cosh y + i \cos x \sinh y.$$

Boundary lines.

$$\sin\left(\pm \frac{\pi}{2} + iy\right) = \pm \cosh y \in \mathbb{R}, \quad |\cosh y| \geq 1.$$

Hence the boundaries map onto the real rays $[1, \infty)$ and $(-\infty, -1]$.

Conformality. $\cos z \neq 0$ on $|x| < \pi/2$ (zeros lie on $x = \pi/2 + k\pi$), so $\sin$ is conformal and injective on the strip.

Image. The image is the simply connected domain whose boundary is those rays; by symmetry (the line $x = 0$ maps to the entire imaginary axis),

$$\boxed{\sin : \{-\frac{\pi}{2} < \Re z < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).}$$

(c) Strip $-\pi < y < \pi$ **under** $\zeta = f(z) = z + e^z$

Let $z = x + iy$. Then

$$f(z) = (x + e^x \cos y) + i(y + e^x \sin y).$$

Critical points (hint).

$$f'(z) = 1 + e^z = 0 \iff e^z = -1 \iff z = i(2k + 1)\pi.$$

There are no critical points in the open strip $-\pi < y < \pi$; the boundary points $z = \pm i\pi$ are the only adjacent critical points.

Boundary images. For $y = \pm\pi$, $\sin y = 0$, $\cos y = -1$:

$$f(x \pm i\pi) = (x - e^x) \pm i\pi.$$

Here $u(x) = x - e^x$ has a unique maximum -1 at $x = 0$ and $u(x) \rightarrow -\infty$ as $x \rightarrow \pm\infty$. Thus each boundary maps to a horizontal half-line

$$(-\infty, -1] + i\pi, \quad (-\infty, -1] - i\pi,$$

meeting at the critical values $-1 \pm i\pi$.

Interior and image. With no interior critical points, f is conformal on the strip. For fixed $y \in (-\pi, \pi)$, $f(x + iy) \rightarrow -\infty + iy$ as $x \rightarrow -\infty$ and $f(x + iy) \sim e^x e^{iy} \rightarrow \infty$ along the ray $\arg \zeta = y$ as $x \rightarrow +\infty$. Hence the image is the plane with the two horizontal half-lines removed:

$$\mathbb{C} \setminus \left((-\infty, -1] + i\pi \cup (-\infty, -1] - i\pi \right).$$

5 Problem 6 — Conformal mappings

Exact problem statement

(a) D is the region exterior to the two circles $|z - 1| = 1$ and $|z + 1| = 1$. Find a conformal mapping from D to the exterior of the unit circle.

Hint: First apply an inversion with respect to the origin ($z \mapsto 1/z$), then rotate and scale so as to use the exponential function to get to a half plane, from where use Möbius.

(b) Find a conformal map from the unit disc $|z| < 1$ onto the strip $-\pi/2 < \eta < \pi/2$, taking the origin to the origin, 1 to $\xi = +\infty$, -1 to $\xi = -\infty$. (Thus, the upper half of the unit circle maps to $\eta = \pi/2$ and the lower half to $\eta = -\pi/2$.)

(c) Find a conformal map of the quarter-disc $0 < |z| < 1$, $0 < \arg z < \pi/2$ to the upper half-plane $\eta > 0$, taking 0 to 0, 1 to 1, and i to ∞ .

Solutions (step by step)

(a) **Exterior of the two circles $\rightarrow$ exterior of $|\zeta| = 1$.**

- **Invert.** Let $w = 1/z$. Then the circles through 0 become lines:

$$|z - 1| = 1 \Rightarrow \Re w = \frac{1}{2}, \quad |z + 1| = 1 \Rightarrow \Re w = -\frac{1}{2}.$$

Hence D maps to the vertical strip $S = \{-\frac{1}{2} < \Re w < \frac{1}{2}\}$.

- **Rotate/scale to a horizontal strip.** Put $s = i\pi w$. Then $|\Im s| < \pi/2$.
- **Exponential to a half-plane.** Let $u = e^s$. Then $|\Im s| < \pi/2 \implies \Re u > 0$.
- **Möbius to exterior unit disk.** Use

$$\Phi(u) = \frac{u + 1}{u - 1}, \quad \Re u > 0 \mapsto |\zeta| > 1.$$

Therefore a concrete map is

$$\zeta = \Phi(e^{i\pi/z}) = \frac{e^{i\pi/z} + 1}{e^{i\pi/z} - 1}.$$

(b) **Unit disk $\rightarrow$ strip $-\pi/2 < \Im \zeta < \pi/2$.**

- **Disk to right half-plane (Cayley).**

$$w = \frac{1+z}{1-z}, \quad |z| < 1 \mapsto \Re w > 0,$$

and $z = 0 \mapsto w = 1$, $z = 1 \mapsto w = \infty$, $z = -1 \mapsto w = 0$.

- **Log to a horizontal strip.**

$$\zeta = \log w \quad (\text{principal branch}), \quad \Im \zeta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Then $\zeta(0) = 0$, $\zeta(1) = +\infty$, $\zeta(-1) = -\infty$.

Hence

$$\boxed{\zeta = \log\left(\frac{1+z}{1-z}\right)}.$$

(c) **Quarter-disk $\rightarrow$ upper half-plane; $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$.**

- **Square to upper semicircle.**

$$t = z^2 : \quad \{0 < |z| < 1, 0 < \arg z < \frac{\pi}{2}\} \mapsto \{|t| < 1, \Im t > 0\}.$$

Moreover $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto -1$.

- **Real Möbius to upper half-plane.** Find $M(t)$ with $M(-1) = \infty$, $M(0) = 0$, $M(1) = 1$.
A suitable choice is

$$M(t) = \frac{2t}{t+1}.$$

Therefore

$$\boxed{\zeta = M(z^2) = \frac{2z^2}{z^2+1}}.$$

It maps the quarter-disk conformally onto $\Im \zeta > 0$ and has $\zeta(0) = 0$, $\zeta(1) = 1$, $\zeta(i) = \infty$.

6(c) Quarter-disk $0 < |z| < 1$, $0 < \arg z < \frac{\pi}{2} \rightarrow$ upper half-plane $\Im \zeta > 0$, with $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$

Goal

Construct an explicit conformal map $\zeta = \zeta(z)$ that maps the open quarter-disk

$$Q = \{z : 0 < |z| < 1, 0 < \arg z < \frac{\pi}{2}\}$$

onto the upper half-plane $\mathbb{H} = \{\zeta : \Im \zeta > 0\}$, sending the three marked boundary points

$$z = 0 \mapsto \zeta = 0, \quad z = 1 \mapsto \zeta = 1, \quad z = i \mapsto \zeta = \infty.$$

Step 1: Square to unfold the angle

Let

$$t = z^2.$$

Write $z = re^{i\theta}$ with $0 < r < 1$ and $0 < \theta < \frac{\pi}{2}$. Then

$$t = r^2 e^{i(2\theta)} \Rightarrow |t| < 1, \quad 0 < \arg t < \pi \iff \Im t > 0.$$

Therefore the squaring map $z \mapsto t = z^2$ sends Q *bijectively and conformally* onto the *open upper semicircle*

$$S = \{t : |t| < 1, \Im t > 0\}.$$

On the boundary rays/arcs we have

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad i \mapsto -1.$$

(Conformality is immediate, since $(z^2)' = 2z \neq 0$ on Q .)

Step 2: Real Möbius map fixing the three marked points and sending S to $\mathbb{H}$

Seek a real-coefficient Möbius transformation

$$M(t) = \frac{at + b}{ct + d}, \quad a, b, c, d \in \mathbb{R}, \quad ad - bc \neq 0,$$

such that

$$M(-1) = \infty, \quad M(0) = 0, \quad M(1) = 1.$$

Imposing the conditions:

- $M(-1) = \infty \Rightarrow c(-1) + d = 0 \Rightarrow d = c.$
- $M(0) = 0 \Rightarrow b/d = 0 \Rightarrow b = 0.$
- $M(1) = 1 \Rightarrow a/(c + d) = 1 \Rightarrow a = c + d = 2c.$

Up to an overall nonzero constant (which does not change the map), we can take $c = 1$ and obtain

$$\boxed{M(t) = \frac{2t}{t + 1}}.$$

Why M maps S into $\Im \zeta > 0$ (and in fact maps the whole upper half-plane to itself).

Let $t = x + iy$ with $y > 0$. Then

$$M(t) = \frac{2t}{1+t}, \quad \Im M(t) = \frac{2\Im(t(1+\bar{t}))}{|1+t|^2} = \frac{2\Im t}{|1+t|^2} = \frac{2y}{|1+t|^2} > 0.$$

Hence M maps $\{\Im t > 0\}$ into itself; in particular it maps the upper semicircle S (which lies in $\Im t > 0$) into the upper half-plane. Moreover

$$M'(t) = \frac{2}{(1+t)^2} \neq 0 \quad \text{on } S,$$

so M is conformal (locally injective) there. The three-point normalization guarantees that it sends the three distinguished boundary points as prescribed:

$$M(-1) = \infty, \quad M(0) = 0, \quad M(1) = 1.$$

Step 3: Compose and verify

Define the desired map

$$\zeta(z) = M(z^2) = \frac{2z^2}{z^2 + 1}.$$

Then the normalization follows immediately:

$$\zeta(0) = 0, \quad \zeta(1) = 1, \quad \zeta(i) = \infty.$$

Because $t = z^2$ has $\Im t > 0$ on Q and $\Im M(t) > 0$ for $\Im t > 0$, we have

$$\Im \zeta(z) = \Im(M(z^2)) > 0 \quad \text{for all } z \in Q.$$

Also $\zeta'(z) = M'(z^2) \cdot 2z = \frac{4z}{(1+z^2)^2} \neq 0$ on Q , hence the map is conformal throughout the domain.

Boundary correspondence (for intuition)

- The *real radius* segment $\{0 < z < 1\}$ maps by $t = z^2$ to $(0, 1)$ and then by M to $(0, 1) \subset \mathbb{R}$; thus that edge lands on the real axis between 0 and 1.
- The *imaginary radius* segment $\{0 < iz < i\}$ maps by $t = z^2$ to $(0, -1)$ and then by M to $(0, \infty)$ on $\mathbb{R}$; thus the edge $0 \rightarrow i$ lands on the ray $[0, \infty)$.
- The *circular arc* $\{e^{i\theta} : 0 < \theta < \frac{\pi}{2}\}$ maps by $t = z^2$ to the unit semicircle $\{e^{i\phi} : 0 < \phi < \pi\}$ and then by M to the real line as a boundary set.

The interior of the quarter-disk maps onto $\Im \zeta > 0$ by the positivity computation for $\Im M(t)$.

Conclusion

The composition

$$\zeta(z) = \frac{2z^2}{z^2 + 1}$$

is a conformal bijection from the open quarter-disk Q onto the upper half-plane $\mathbb{H}$, satisfying the three mapping conditions $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$.

(b) Unit disk $\rightarrow$ strip $-\frac{\pi}{2} < \Im \zeta < \frac{\pi}{2}$

Disk to right half-plane (Cayley).

$$w = \frac{1+z}{1-z}, \quad |z| < 1 \implies \operatorname{Re} w > 0,$$

and on the marked points

$$z = 0 \mapsto w = 1, \quad z = 1 \mapsto w = \infty, \quad z = -1 \mapsto w = 0.$$

Moreover, the unit circle $|z| = 1 \setminus \{1\}$ maps to the imaginary axis $\operatorname{Re} w = 0$.

Log to a horizontal strip. Use the principal logarithm on the right half-plane:

$$\zeta = \log w = \ln |w| + i \arg(w), \quad \arg(w) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

so

$$-\frac{\pi}{2} < \Im \zeta < \frac{\pi}{2}.$$

At the marked points:

$$\zeta(0) = \log 1 = 0, \quad \zeta(1) = \log(\infty) = +\infty, \quad \zeta(-1) = \log(0) = -\infty.$$

Which boundary goes where. For $z = e^{i\theta}$,

$$w = \frac{1 + e^{i\theta}}{1 - e^{i\theta}} = -i \cot\left(\frac{\theta}{2}\right) \in i\mathbb{R}.$$

Hence

$$0 < \theta < \pi \text{ (upper semicircle)} \Rightarrow \Im w < 0 \Rightarrow \Im \zeta = -\frac{\pi}{2},$$

$$\pi < \theta < 2\pi \text{ (lower semicircle)} \Rightarrow \Im w > 0 \Rightarrow \Im \zeta = +\frac{\pi}{2}.$$

(If one wants the upper semicircle to land on $\Im \zeta = +\pi/2$, use $\zeta = -\log\left(\frac{1+z}{1-z}\right)$.)

Hence

$$\boxed{\zeta = \log\left(\frac{1+z}{1-z}\right)}$$

is a conformal bijection from $|z| < 1$ onto the strip $-\pi/2 < \Im \zeta < \pi/2$ with $0 \mapsto 0$, $1 \mapsto +\infty$, $-1 \mapsto -\infty$.

(a) Exterior of the two circles $|z - 1| = 1$ and $|z + 1| = 1$ $\rightarrow$ exterior of $|\zeta| = 1$

Goal. Let

$$D = \{z \in \mathbb{C} : |z - 1| \geq 1, |z + 1| \geq 1\} \setminus (\{|z - 1| = 1\} \cup \{|z + 1| = 1\}),$$

the region *exterior to both* unit circles centered at ± 1 . Construct an explicit conformal bijection $F : D \rightarrow \{\zeta : |\zeta| > 1\}$.

Step 1: Inversion. Let $w = \frac{1}{z}$. For any $a \in \mathbb{C}$, the circle $\{|z - a| = |a|\}$ (which passes through 0) satisfies

$$|z - a| = |a| \iff \left|\frac{1}{w} - a\right| = |a| \iff |1 - aw| = |a||w| \iff 1 = 2\Re(aw).$$

Hence circles through 0 invert to *lines*. For $a = 1$ and $a = -1$ we obtain

$$|z - 1| = 1 \iff \Re w = \frac{1}{2}, \quad |z + 1| = 1 \iff \Re w = -\frac{1}{2}.$$

Which side is the exterior? Using $|1 - aw|^2 = (1 - 2\Re(aw) + |a|^2|w|^2)$ we compute

$$|z - 1| > 1 \iff |1 - w| > |w| \iff 1 - 2\Re w > 0 \iff \Re w < \frac{1}{2},$$

$$|z + 1| > 1 \iff |1 + w| > |w| \iff 1 + 2\Re w > 0 \iff \Re w > -\frac{1}{2}.$$

Therefore the inversion $w = 1/z$ sends D bijectively onto the *vertical strip*

$$S = \left\{ w \in \mathbb{C} : -\frac{1}{2} < \Re w < \frac{1}{2} \right\}.$$

(Since $z \neq 0$ on D , the inversion is analytic and injective there.)

Step 2: Rotate/scale to a horizontal strip. Define

$$s = i\pi w.$$

Write $w = u + iv$ with $-\frac{1}{2} < u < \frac{1}{2}$. Then $s = i\pi(u + iv) = -\pi v + i\pi u$, so

$$|\Im s| = |\pi u| < \frac{\pi}{2}.$$

Consequently S is mapped conformally onto the horizontal strip

$$T = \left\{ s \in \mathbb{C} : |\Im s| < \frac{\pi}{2} \right\}.$$

Step 3: Exponential to the right half-plane. Set

$$u = e^s.$$

If $s = \sigma + i\tau$ with $|\tau| < \pi/2$, then

$$\Re u = \Re(e^\sigma(\cos \tau + i \sin \tau)) = e^\sigma \cos \tau > 0,$$

because $\cos \tau > 0$ on $(-\pi/2, \pi/2)$. Hence $e^{(\cdot)}$ maps T conformally and bijectively onto the *right half-plane*

$$H = \{ u \in \mathbb{C} : \Re u > 0 \},$$

and the boundary lines $\Im s = \pm\pi/2$ go to the imaginary axis $\Re u = 0$.

Step 4: Möbius map from the right half-plane to the exterior disk. Consider

$$\Phi(u) = \frac{u + 1}{u - 1}.$$

For $u \in H$ we have the equivalence

$$|\Phi(u)| > 1 \iff |u + 1| > |u - 1| \iff (u + 1)(\bar{u} + 1) > (u - 1)(\bar{u} - 1) \iff 4\Re u > 0,$$

which holds precisely on H . Thus Φ maps H conformally onto $\{\zeta : |\zeta| > 1\}$ and sends the imaginary axis to the unit circle $|\zeta| = 1$.

Step 5: Composition and explicit formula. The composition of the four conformal bijections yields

$$F : D \xrightarrow{z \mapsto w=1/z} S \xrightarrow{w \mapsto s=i\pi w} T \xrightarrow{s \mapsto u=e^s} H \xrightarrow{u \mapsto \Phi(u)} \{|\zeta| > 1\}.$$

Therefore an explicit mapping is

$$\boxed{\zeta = F(z) = \Phi(e^{i\pi/z}) = \frac{e^{i\pi/z} + 1}{e^{i\pi/z} - 1}}.$$

Every factor is analytic and injective on the relevant domain; hence F is conformal and bijective from D onto $\{|\zeta| > 1\}$.

Asymptotic check at infinity (sanity). As $z \rightarrow \infty$ we have $e^{i\pi/z} = 1 + \frac{i\pi}{z} + O(z^{-2})$, so

$$\zeta = \frac{2 + \frac{i\pi}{z} + O(z^{-2})}{\frac{i\pi}{z} + O(z^{-2})} = \frac{2}{i\pi} z \left(1 + O(z^{-1})\right) = -\frac{2i}{\pi} z \left(1 + O(z^{-1})\right),$$

which confirms that far out in D the map behaves like an affine scaling of z , as expected.

Boundary correspondence (orientation).

- The line $\Re w = \frac{1}{2}$ (image of $|z - 1| = 1$) becomes $\Im s = \frac{\pi}{2}$, then the imaginary axis, and finally the unit circle $|\zeta| = 1$.
- The line $\Re w = -\frac{1}{2}$ (image of $|z + 1| = 1$) becomes $\Im s = -\frac{\pi}{2}$, then the same imaginary axis, and again $|\zeta| = 1$.
- The interior $-\frac{1}{2} < \Re w < \frac{1}{2}$ maps to $|\Im s| < \frac{\pi}{2}$, then to $\Re u > 0$, and finally to $|\zeta| > 1$.

This completes the construction and detailed verification.

6 Problem 7: Temperature in a right half-plane with a removed disk

Exact statement (verbatim)

The domain D consists of the right-hand half plane $x > 0$ with the circle $|z - a| = b$, $0 < b < a$, and its interior removed. Find the temperature $u(x, y)$ in steady heat flow if $u = 0$ on the y axis, $u = 1$ on $|z - a| = b$, and $u \rightarrow 0$ at infinity.

Hint: Show that the mapping $\zeta = \frac{z - \alpha}{z + \alpha}$, with α real and positive, takes D onto an annular region with the imaginary axis mapping to $|\zeta| = 1$ and show that, if $\alpha^2 = a^2 - b^2$, then the image of D is a concentric circular annulus.

Background: why the suggested Möbius map produces an annulus

Consider the real-parameter Möbius map

$$\zeta = \zeta(z) = \frac{z - \alpha}{z + \alpha}, \quad \alpha > 0.$$

Imaginary axis. For $z = iy$ we have

$$\left| \frac{iy - \alpha}{iy + \alpha} \right| = 1 \quad (\text{numerator and denominator are conjugates}),$$

so the y -axis maps to the unit circle $|\zeta| = 1$.

Image of the circle $|z - a| = b$. Write $z = a + be^{i\theta}$ and compute

$$|\zeta|^2 = \frac{|a - \alpha + be^{i\theta}|^2}{|a + \alpha + be^{i\theta}|^2} = \frac{A + B \cos \theta}{C + D \cos \theta},$$

where

$$A = (a - \alpha)^2 + b^2, \quad B = 2b(a - \alpha), \quad C = (a + \alpha)^2 + b^2, \quad D = 2b(a + \alpha).$$

For $|\zeta|$ to be constant on the circle (i.e. the image is a circle *centered at the origin*), the $\cos \theta$ -dependence must drop out:

$$\frac{A + B \cos \theta}{C + D \cos \theta} \equiv \frac{A}{C} \iff AD = BC.$$

We now verify that

$$AD - BC = 2b \left[(a + \alpha)((a - \alpha)^2 + b^2) - (a - \alpha)((a + \alpha)^2 + b^2) \right] = 4b\alpha(\alpha^2 + b^2 - a^2).$$

Thus $AD = BC$ is equivalent to

$$\boxed{\alpha^2 = a^2 - b^2}.$$

With this choice (note $a > b > 0 \Rightarrow \alpha > 0$), the ratio becomes *constant*:

$$|\zeta|^2 = \frac{A}{C} = \frac{(a - \alpha)^2 + b^2}{(a + \alpha)^2 + b^2} = \frac{2a(a - \alpha)}{2a(a + \alpha)} = \frac{a - \alpha}{a + \alpha}.$$

Hence the circle $|z - a| = b$ maps to the circle $|\zeta| = \rho$ with

$$\boxed{\rho = \sqrt{\frac{a - \alpha}{a + \alpha}} \in (0, 1).}$$

Image of the domain. Therefore, for $\alpha = \sqrt{a^2 - b^2}$, the map $z \mapsto \zeta$ sends the domain D (right half-plane with the closed disk $\{|z - a| \leq b\}$ removed) onto the concentric circular *annulus*

$$\mathcal{A} = \{ \rho < |\zeta| < 1 \}.$$

The boundary pieces correspond as follows:

$$\{x = 0\} \mapsto |\zeta| = 1, \quad \{|z - a| = b\} \mapsto |\zeta| = \rho.$$

Solving the Dirichlet problem on the annulus

Let $U(\zeta)$ denote the temperature in the ζ -plane. Harmonicity is preserved by conformal maps, so U is harmonic on $\mathcal{A}$ and the boundary data become

$$U = 0 \quad \text{on} \quad |\zeta| = 1, \quad U = 1 \quad \text{on} \quad |\zeta| = \rho.$$

By rotational symmetry, U depends only on $r = |\zeta|$. The general radial harmonic function on an annulus is $A \log r + B$. Imposing the boundary values gives

$$U(r) = \frac{\log(1/r)}{\log(1/\rho)} = \frac{-\log r}{-\log \rho} = \frac{\log r}{\log \rho} \quad \text{for} \quad \rho < r < 1.$$

(Any of the three displayed forms is correct; we will use the first one momentarily.)

Pullback to the original z -plane and final formula

The physical temperature is $u(z) = U(\zeta(z))$ with $r = |\zeta(z)| = \left| \frac{z-\alpha}{z+\alpha} \right|$. Hence

$$u(z) = U\left(\left|\frac{z-\alpha}{z+\alpha}\right|\right) = \frac{\log\left(\frac{1}{\left|\frac{z-\alpha}{z+\alpha}\right|}\right)}{\log\left(\frac{1}{\rho}\right)} = \frac{\log\left|\frac{z+\alpha}{z-\alpha}\right|}{\log\left(\frac{1}{\rho}\right)}.$$

Since $\rho = \sqrt{\frac{a-\alpha}{a+\alpha}}$ we have

$$\log\left(\frac{1}{\rho}\right) = \frac{1}{2} \log\left(\frac{a+\alpha}{a-\alpha}\right).$$

Therefore a convenient explicit form is

$$u(z) = \frac{2 \log\left|\frac{z+\alpha}{z-\alpha}\right|}{\log\left(\frac{a+\alpha}{a-\alpha}\right)}, \quad \alpha = \sqrt{a^2 - b^2}.$$

Checks of the boundary conditions and far field

- **On the y -axis $x = 0$:** for $z = iy$ we have $\left|\frac{z-\alpha}{z+\alpha}\right| = 1$, so the numerator $\log\left|\frac{z+\alpha}{z-\alpha}\right| = 0$ and hence $u = 0$.
 - **On the circle $|z - a| = b$:** by construction $|\zeta| = \rho$, i.e. $\left|\frac{z-\alpha}{z+\alpha}\right| = \rho$, so $\log\left|\frac{z+\alpha}{z-\alpha}\right| = \log\left(\frac{1}{\rho}\right)$ and the fraction evaluates to $u = 1$.
 - **At infinity:** as $|z| \rightarrow \infty$, $\left|\frac{z+\alpha}{z-\alpha}\right| \rightarrow 1$, so the numerator $\rightarrow 0$ and $u(z) \rightarrow 0$.
-

Summary

With $\alpha = \sqrt{a^2 - b^2}$ and $\rho = \sqrt{\frac{a - \alpha}{a + \alpha}}$, the Möbius map $\zeta = \frac{z - \alpha}{z + \alpha}$ carries D onto the annulus $\{\rho < |\zeta| < 1\}$. The unique harmonic function with $U = 1$ on $|\zeta| = \rho$ and $U = 0$ on $|\zeta| = 1$ is $U(r) = \log(1/r)/\log(1/\rho)$, and hence

$$u(z) = \frac{2 \log \left| \frac{z + \alpha}{z - \alpha} \right|}{\log \left(\frac{a + \alpha}{a - \alpha} \right)}.$$

This u satisfies all the prescribed boundary conditions and $u(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

Hyperbolic maps: $\sinh$, $\cosh$, $\tanh$, $\coth$ — background, zeros/poles, and strip mappings

1. $w = \sinh z$

Background: entire, odd; period $2\pi i$. **Zeros:** $z = i\pi k$; **poles:** none.

Decomposition: $w = u + iv = \sinh x \cos y + i \cosh x \sin y$.

Vertical lines $x = \text{const}$: $\left(\frac{u}{\sinh x}\right)^2 + \left(\frac{v}{\cosh x}\right)^2 = 1$ (ellipses, foci $\pm i$).

Horizontal lines $y = \text{const}$: $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = -1$ (hyperbolas, foci $\pm i$).

Strip mapping:

$$\sinh : \{|\Im z| < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty))$$

Boundaries $y = \pm \frac{\pi}{2}$ map to imaginary rays $\pm i[1, \infty)$.

2. $w = \cosh z$

Background: entire, even; period $2\pi i$. **Zeros:** $z = i\pi(k + \frac{1}{2})$; **poles:** none.

Decomposition: $w = u + iv = \cosh x \cos y + i \sinh x \sin y$.

Vertical lines $x = \text{const}$: $\left(\frac{u}{\cosh x}\right)^2 + \left(\frac{v}{\sinh x}\right)^2 = 1$ (ellipses, foci ± 1).

Horizontal lines $y = \text{const}$: $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = 1$ (hyperbolas, foci ± 1).

Strip mapping:

$$\cosh : \{|\Im z| < \pi\} \xrightarrow{2-1} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$$

On the half-strip $0 < |\Im z| < \pi$, the map is 1-1 onto the slit plane.

3. $w = \tanh z$

Background: meromorphic, odd; period $i\pi$. **Zeros:** $z = i\pi k$; **poles:** $z = i\pi(k + \frac{1}{2})$.

Strip $\rightarrow$ disk:

$$\tanh : \{|\Im z| < \frac{\pi}{2}\} \xrightarrow{1-1} \mathbb{D}$$

Real line maps to $(-1, 1)$; as $|\Im z| \rightarrow \pi/2$, $|\tanh z| \rightarrow 1$.

4. $w = \coth z$

Background: meromorphic, odd; period $i\pi$. **Zeros:** $z = i\pi(k + \frac{1}{2})$; **poles:** $z = i\pi k$.

Strip $\rightarrow$ exterior disk:

$$\coth : \{|\Im z| < \frac{\pi}{2}\} \setminus \{0\} \xrightarrow{1-1} \{|w| > 1\}$$

Real line maps to $(-\infty, -1) \cup (1, \infty)$; $|\Im z| \rightarrow \pi/2$ gives $|w| \rightarrow 1^+$.

Hyperbolic Conformal Maps: $\sinh$, $\cosh$, $\tanh$, $\coth$

Notation and preliminaries

Throughout, write $z = x + iy$, $x, y \in \mathbb{R}$, and $w = u + iv$ for the image. Recall the basic identities

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}.$$

We use the principal branch of the logarithm $\log$ on $\mathbb{C} \setminus (-\infty, 0]$, and principal square roots $\sqrt{\cdot}$ with branch cut on $(-\infty, 0]$ unless stated otherwise.

1. $w = \sinh z$

Analytic background.

$\sinh$ is an *entire*, *odd* function, with imaginary-periodicity:

$$\sinh(-z) = -\sinh z, \quad \sinh(z + 2\pi i) = \sinh z.$$

Zeros occur at $z_k = i\pi k$ for $k \in \mathbb{Z}$ (all simple). No poles.

Cauchy–Riemann decomposition.

With $z = x + iy$,

$$\sinh z = \sinh x \cos y + i \cosh x \sin y \quad \Rightarrow \quad u = \sinh x \cos y, \quad v = \cosh x \sin y.$$

From this, one obtains orthogonal families of confocal conics:

$$\begin{aligned} x = \text{const} : \quad & \left(\frac{u}{\sinh x} \right)^2 + \left(\frac{v}{\cosh x} \right)^2 = 1 && (\text{ellipses, foci at } \pm i), \\ y = \text{const} : \quad & \left(\frac{u}{\cos y} \right)^2 - \left(\frac{v}{\sin y} \right)^2 = -1 && (\text{hyperbolas, foci at } \pm i). \end{aligned}$$

Real and imaginary axes map as:

$$y = 0 : w = \sinh x \in \mathbb{R} \text{ (onto } \mathbb{R}), \quad x = 0 : w = i \sin y \text{ (onto the segment } i[-1, 1]).$$

Jacobian and conformality.

$\sinh'(z) = \cosh z$, and $\cosh z \neq 0$ on the strip $|\Im z| < \frac{\pi}{2}$; hence $\sinh$ is conformal there.

Canonical strip mapping (global picture).

$\sinh : \{ \Im z < \frac{\pi}{2} \} \xrightarrow{\text{biholomorphism}} \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)) .$
--

Explanation. The boundary lines $y = \pm \frac{\pi}{2}$ map to the imaginary rays $\pm i[1, \infty)$ (via $v = \cosh x \sin y$ with $\sin(\pm\pi/2) = \pm 1$ and $u = \sinh x \cos(\pm\pi/2) = 0$), while the interior maps injectively since $\cosh z \neq 0$ and periodicity in the imaginary direction has period 2π , not π .

Inverse branch and branch cuts.

A principal inverse on the slit domain is

$$\operatorname{arsinh} w = \log(w + \sqrt{w^2 + 1}),$$

with branch cuts along $i(-\infty, -1] \cup i[1, \infty)$ so that $\Im(\operatorname{arsinh} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Useful identities.

$$\cosh^2 z - \sinh^2 z = 1, \quad \sinh(z + \zeta) = \sinh z \cosh \zeta + \cosh z \sinh \zeta.$$

2. $w = \cosh z$

Analytic background.

$\cosh$ is *entire*, *even*, and $2\pi i$ -periodic:

$$\cosh(-z) = \cosh z, \quad \cosh(z + 2\pi i) = \cosh z.$$

Zeros at $z = i\pi(k + \frac{1}{2})$, all simple. No poles.

Cauchy–Riemann decomposition.

With $z = x + iy$,

$$\cosh z = \cosh x \cos y + i \sinh x \sin y \quad \Rightarrow \quad u = \cosh x \cos y, \quad v = \sinh x \sin y.$$

Again one gets confocal conics:

$$\begin{aligned} x = \text{const} : \quad & \left(\frac{u}{\cosh x} \right)^2 + \left(\frac{v}{\sinh x} \right)^2 = 1 && (\text{ellipses, foci at } \pm 1), \\ y = \text{const} : \quad & \left(\frac{u}{\cos y} \right)^2 - \left(\frac{v}{\sin y} \right)^2 = 1 && (\text{hyperbolas, foci at } \pm 1). \end{aligned}$$

On the axes,

$$y = 0 : w = \cosh x \in [1, \infty), \quad x = 0 : w = \cos y \in [-1, 1].$$

Critical points and injectivity on strips.

$\cosh'(z) = \sinh z$, which vanishes at $z = i\pi k$. Thus $\cosh$ fails to be injective across lines $y = k\pi$, but is injective on any strip avoiding these lines (e.g. $0 < \Im z < \pi$).

Canonical strip mapping.

$\cosh : \{ 0 < \Im z < \pi \} \xrightarrow{\text{biholomorphism}} \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)) .$

Explanation. The midline $y = 0$ maps to $[1, \infty)$; the top boundary $y = \pi$ maps to $(-\infty, -1]$; injectivity holds on $0 < \Im z < \pi$ since there $\sinh z \neq 0$.

Inverse branch and branch cuts.

$$\operatorname{arcosh} w = \log \left(w + \sqrt{w-1} \sqrt{w+1} \right),$$

with branch cuts along $(-\infty, -1] \cup [1, \infty)$ so that $\Re(\operatorname{arcosh} w) \geq 0$ and $0 < \Im(\operatorname{arcosh} w) < \pi$ on the principal sheet.

Useful identities.

$$\cosh(z + \zeta) = \cosh z \cosh \zeta + \sinh z \sinh \zeta, \quad \cosh z = 1 + 2 \sinh^2 \frac{z}{2}.$$

3. $w = \tanh z$ **Analytic background.**

$\tanh$ is *meromorphic*, *odd*, and $i\pi$ -periodic:

$$\tanh(-z) = -\tanh z, \quad \tanh(z + i\pi) = \tanh z.$$

Zeros at $z = i\pi k$; poles (simple) at $z = i\pi(k + \frac{1}{2})$.

Strip $\leftrightarrow$ disk (Cayley-type).

$$\boxed{\tanh : \{ |\Im z| < \frac{\pi}{2} \} \xrightarrow{\text{biholomorphism}} \mathbb{D} .}$$

Boundary correspondence. The real axis $y = 0$ maps bijectively onto $(-1, 1)$ (strictly increasing). As $y \rightarrow \pm \frac{\pi}{2}$, $|\tanh(x + iy)| \rightarrow 1$ uniformly in x ; hence the horizontal boundaries map to $\partial\mathbb{D}$.

Inverse (disk $\rightarrow$ strip).

$$\operatorname{artanh} w = \frac{1}{2} \log \left(\frac{1+w}{1-w} \right), \quad |w| < 1,$$

which maps $\mathbb{D}$ conformally onto $\{ |\Im z| < \frac{\pi}{2} \}$.

Derivative and Schwarz–Pick.

$$(\tanh z)' = \operatorname{sech}^2 z, \quad |\tanh'(z)| = \frac{1}{|\cosh z|^2}$$

and conjugating by a disk automorphism shows $\tanh$ is extremal for the disk-strip version of Schwarz–Pick.

Level sets.

Lines $x = \text{const}$ and $y = \text{const}$ in the strip map to orthogonal circle arcs inside $\mathbb{D}$ meeting $\partial\mathbb{D}$ orthogonally (images of preimages under the conformal equivalence with the half-plane followed by a Cayley map).

4. $w = \coth z$

Analytic background.

$\coth$ is *meromorphic*, *odd*, and $i\pi$ -periodic:

$$\coth(-z) = -\coth z, \quad \coth(z + i\pi) = \coth z.$$

Poles (simple) at $z = i\pi k$; zeros (simple) at $z = i\pi(k + \frac{1}{2})$.

Strip $\leftrightarrow$ exterior disk.

$$\boxed{\coth : \{ |\Im z| < \frac{\pi}{2} \} \setminus \{0\} \xrightarrow{\text{biholomorphism}} \{ |w| > 1 \} .}$$

Boundary correspondence. The real axis (excluding 0) maps to the rays $(-\infty, -1) \cup (1, \infty)$. As $y \rightarrow \pm \frac{\pi}{2}$, $|\coth(x + iy)| \rightarrow 1^+$; thus horizontal boundaries map to the unit circle from the *exterior* side.

Inverse (exterior disk $\rightarrow$ strip).

$$\operatorname{arccoth} w = \frac{1}{2} \log \left(\frac{w+1}{w-1} \right), \quad |w| > 1,$$

with the standard logarithm branch so that $\Im(\operatorname{arccoth} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Relation with $\tanh$.

$$\coth z = \tanh \left(z + \frac{i\pi}{2} \right),$$

so the exterior-disk picture is just the disk picture shifted vertically by $i\pi/2$ and inverted via $w \mapsto 1/w$.

5. Global relations, inverse formulas, and composition patterns

Inverse formulas (principal branches).

$$\begin{aligned} \operatorname{arsinh} w &= \log(w + \sqrt{w^2 + 1}), & \operatorname{arcosh} w &= \log(w + \sqrt{w-1} \sqrt{w+1}), \\ \operatorname{artanh} w &= \frac{1}{2} \log \left(\frac{1+w}{1-w} \right), & \operatorname{arccoth} w &= \frac{1}{2} \log \left(\frac{w+1}{w-1} \right). \end{aligned}$$

Branch cuts are chosen to match the canonical target domains described above.

Disk/half-plane equivalences via Cayley.

Using the Cayley map $C(\zeta) = \frac{1+\zeta}{1-\zeta}$ and its inverse $C^{-1}(w) = \frac{w-1}{w+1}$,

$$\tanh z = C^{-1}(e^{2z}), \quad \coth z = C^{-1}(e^{2z})^{-1}, \quad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}.$$

Thus strip $\leftrightarrow$ disk/exterior-disk mappings for $\tanh$ and $\coth$ are just the exponential's strip-to-sector equivalences composed with a Cayley map.

Growth and boundary behavior.

As $x \rightarrow \pm\infty$ with y fixed, $\sinh z \sim \frac{1}{2}e^x e^{iy}$ and $\cosh z \sim \frac{1}{2}e^x e^{iy}$; hence horizontal lines map to asymptotically radial curves. For $\tanh$ and $\coth$, $|\tanh(x+iy)| \rightarrow 1$ and $|\coth(x+iy)| \rightarrow 1$ exponentially in $|x|$ (uniform in y within the strip height).

Critical sets and injectivity regions.

$\sinh'(z) = \cosh z$ vanishes along $y = \frac{\pi}{2} + \pi\mathbb{Z}$; $\cosh'(z) = \sinh z$ vanishes along $y = \pi\mathbb{Z}$. Therefore, the natural injectivity strips are $|\Im z| < \frac{\pi}{2}$ for $\sinh$ and $0 < \Im z < \pi$ for $\cosh$. For $\tanh$ and $\coth$, poles on the horizontal boundaries determine the natural strip height π .

6. Summary: canonical conformal equivalences (boxed)

$$\sinh : \{ |\Im z| < \frac{\pi}{2} \} \longrightarrow \mathbb{C} \setminus i((-\infty, -1] \cup [1, \infty)) \quad (\text{biholomorphism}) .$$

$$\cosh : \{ 0 < \Im z < \pi \} \longrightarrow \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)) \quad (\text{biholomorphism}) .$$

$$\tanh : \{ |\Im z| < \frac{\pi}{2} \} \longrightarrow \mathbb{D} \quad (\text{biholomorphism}) .$$

$$\coth : \{ |\Im z| < \frac{\pi}{2} \} \setminus \{0\} \longrightarrow \{ |w| > 1 \} \quad (\text{biholomorphism}) .$$

7. (Optional) Proof sketches for strip mappings

Injectivity for $\sinh$ on $|\Im z| < \pi/2$. If $\sinh z_1 = \sinh z_2$ then $e^{z_1} - e^{-z_1} = e^{z_2} - e^{-z_2}$. Rearranging gives $\{e^{z_1}, e^{-z_1}\} = \{e^{z_2}, e^{-z_2}\}$. Taking logs in the horizontal strip $|\Im z| < \pi/2$ (where the exponential is injective modulo $2\pi i$) forces $z_1 = z_2$.

Boundary images for $\sinh$. Set $z = x \pm i\pi/2$. Then $\sinh z = \sinh x \cdot 0 \pm i \cosh x$, so the boundaries map to $\pm i[1, \infty)$; the interior cannot reach those rays by the open mapping theorem and the maximum modulus principle applied to $1/(\sinh z \mp i)$.

Analogous arguments apply to $\cosh$ (use $\sinh$ zeros to avoid critical lines) and to $\tanh, \coth$ using the representation via exponentials and the Cayley transform.

Conformal maps: an overview catalogue (no details)

1. Cayley (disk $\leftrightarrow$ right half-plane).

$$w = \frac{1+z}{1-z} \quad \text{inverse:} \quad z = \frac{w-1}{w+1} .$$

2. Half-plane $\rightarrow$ half-plane (three-point normalization).

$$w = \frac{(z-a)(c-b)}{(z-b)(c-a)} .$$

3. Disk automorphisms.

$$\phi_{a,\theta}(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z}, \quad |a| < 1 .$$

4. Strip $|\Im z| < h/2 \rightarrow$ right half-plane.

$$w = \exp\left(\frac{\pi z}{h}\right) .$$

5. **Strip** $|\Im z| < h/2 \rightarrow$ **unit disk**.

$$\zeta = \frac{e^{\pi z/h} - 1}{e^{\pi z/h} + 1}.$$

6. **Strip** $\rightarrow$ **strip (height scaling)**.

$$z \mapsto \frac{h_2}{h_1} z.$$

7. **Band** $a < \Im z < b \rightarrow$ **centered strip**.

$$z \mapsto \frac{\pi}{b-a} \left(z - i \frac{a+b}{2} \right).$$

8. **Sector** $\{0 < \arg z < \alpha\} \rightarrow$ **upper half-plane**.

$$w = z^{\pi/\alpha} \quad (\text{principal branch}).$$

9. **Wedge** $\rightarrow$ **disk**.

Compose item (8) with Cayley (item 1).

10. **Upper half-plane** $\rightarrow$ **disk (prescribed triple)**.

$$\Phi(z) = \frac{(z-a)}{(z-\bar{a})} \cdot \frac{(b-\bar{a})}{(b-a)}.$$

11. **Annulus** $r < |z| < 1 \leftrightarrow$ **strip**.

$$w = \log z \quad \text{maps to} \quad \log r < \Re w < 0.$$

12. **Annulus** $\rightarrow$ **disk**.

Use $w = \log z$ then item (5).

13. **Half-plane minus disk** $\rightarrow$ **annulus**.

$$\zeta = \frac{z-\alpha}{z+\alpha}, \quad \alpha^2 = a^2 - b^2.$$

14. **Disk with off-center hole** $\rightarrow$ **annulus**.

Center via ϕ_a (item 3), then items (11)–(12).

15. **Punctured plane** $\leftrightarrow$ **cylinder**.

$$w = \log z \quad (\text{period } 2\pi i).$$

16. **Strip** $\rightarrow$ **punctured plane**.

$$z \mapsto e^z.$$

17. **Ray-slit plane** $\mathbb{C} \setminus [0, \infty) \rightarrow$ **half-plane**.

$$w = \sqrt{z} \quad (\text{choose branch}).$$

18. **Segment-slit plane** $\mathbb{C} \setminus [a, b] \rightarrow$ **half-plane**.

$$w = \sqrt{\frac{z-a}{z-b}}.$$

19. **Exterior of $[-1, 1]$** $\leftrightarrow$ **exterior unit circle (Joukowski)**.

$$w = z + \frac{1}{z}, \quad z = \frac{w \pm \sqrt{w^2 - 4}}{2}.$$

20. **Exterior of ellipse** $\leftrightarrow$ **exterior unit circle**.

$$w = \frac{1}{2} \left(cz + \frac{1}{cz} \right) \quad (c \text{ from axes}).$$

21. **Right half-plane** $\rightarrow$ **vertical strip**.

$$z \mapsto \frac{1}{\pi} \log \frac{z-1}{z+1}.$$

22. **Upper half-plane** $\rightarrow$ **rectangle (SC/elliptic)**.

$$\zeta = \int^z \frac{dt}{\sqrt{(t-a)(t-b)(t-c)(t-d)}}.$$

23. **Rectangle** $\rightarrow$ **disk (Jacobi sn)**.

$$\zeta = \text{sn} \left(\frac{K(k)}{L} z; k \right).$$

24. **Half-disk** $\leftrightarrow$ **half-plane**.

Restriction of Cayley (item 1) and inverse.

25. **Lens (intersection of two disks)** $\rightarrow$ **disk**.

Möbius normalizes the circles; then a power map; then Cayley.

26. **Half-strip** $\{x > 0, 0 < \Im z < h\}$ $\rightarrow$ **half-disk**.

$$\exp(\pi z/h) \quad \text{then power, then Cayley.}$$

27. **Polygon** $\rightarrow$ **half-plane (Schwarz–Christoffel)**.

$$\Phi'(z) = C \prod_k (z - z_k)^{\alpha_k - 1}, \quad \sum \alpha_k = 2.$$

28. **Upper half-plane** $\rightarrow$ **slit half-plane**.

$$w = z + \frac{1}{z} \quad \text{then a real Möbius adjustment.}$$

29. **Disk** $\rightarrow$ **disk with boundary arc removed**.

Automorphism to put endpoints at ± 1 , then boundary power z^λ .

30. **Finite Blaschke product (disk $\rightarrow$ disk, degree n).**

$$B(z) = e^{i\theta} \prod_{k=1}^n \frac{z - a_k}{1 - \bar{a}_k z}.$$

1. Cayley transform: $\mathbb{D} \leftrightarrow$ right half-plane

Map and inverse.

$$w = \frac{1+z}{1-z} \quad \text{maps} \quad \mathbb{D} = \{|z| < 1\} \quad \text{onto} \quad \{\Re w > 0\}, \quad z = \frac{w-1}{w+1}$$

is the inverse.

Why $\Re w > 0$.

Compute

$$\Re\left(\frac{1+z}{1-z}\right) = \frac{1-|z|^2}{|1-z|^2}.$$

Thus $\Re w > 0$ for $|z| < 1$, and $\Re w = 0$ for $|z| = 1$ (except $z = 1$, which maps to $w = \infty$).
Therefore

$$|z| < 1 \iff \Re w > 0, \quad |z| = 1 \iff \Re w = 0.$$

Boundary correspondence.

$z = 0 \mapsto w = 1$, $z \rightarrow 1^- \mapsto w \rightarrow +\infty$, and $\partial\mathbb{D} \setminus \{1\}$ maps to the line $\Re w = 0$.

Circles/lines orthogonal to $\partial\mathbb{D}$ map to *vertical* lines in the w -plane.

2. Three-point Möbius normalization: half-plane $\leftrightarrow$ half-plane

Cross-ratio map to the real axis.

Let L be a line or circle, and choose three distinct points $a, b, c \in L$. Define

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then $\Phi(L) \subset \mathbb{R} \cup \{\infty\}$, with

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1,$$

and the two sides of L map to the half-planes $\Im \Phi(z) \gtrless 0$.

Half-plane to half-plane via three boundary points.

Given half-planes H_1, H_2 with boundary arcs L_1, L_2 , pick ordered boundary triples $a, b, c \in L_1$ and $A, B, C \in L_2$ (same cyclic order). Then

$$T(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} \cdot \frac{(C-A)}{(C-B)}$$

is a Möbius map sending $L_1 \rightarrow L_2$ with

$$a \mapsto A, \quad b \mapsto B, \quad c \mapsto C,$$

and taking the chosen side of H_1 onto the chosen side of H_2 .

Specialization to the real line.

To send L to $\mathbb{R}$, use Φ above; then pre/postcompose with an *affine real* map

$$\xi \mapsto \alpha \xi + \beta, \quad \alpha > 0,$$

to obtain any target half-plane, e.g. $\Re w > 0$ or $\Im w > 0$.